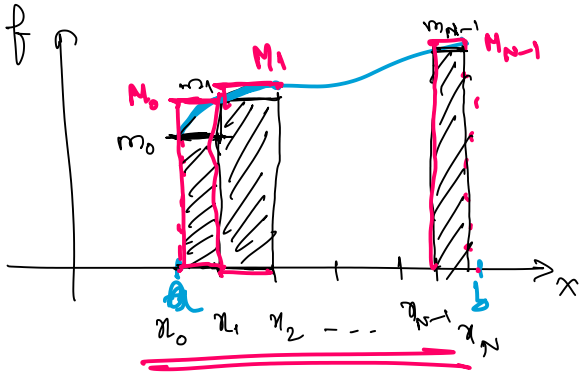
Lecture 16

Tuesday, 15 September 2026 8:25 AM



Meaning of $\int_a^b f(x)\,dx$?

$\sup(\text{lower sums}) = m_0(x_1 - x_0) + m_1(x_2 - x_1) + \cdots + m_{N-1}(x_N - x_{N-1})$

$< \int_a^b f(x)\,dx$ → Area

$< M_0(x_1 - x_0) + M_1(x_2 - x_1) + \cdots + M_{N-1}(x_N - x_{N-1})$

inf upper sums

Def:

$$\int_a^b f(x)\,dx = \sup(\text{lower sums}) = \inf(\text{upper sums})$$

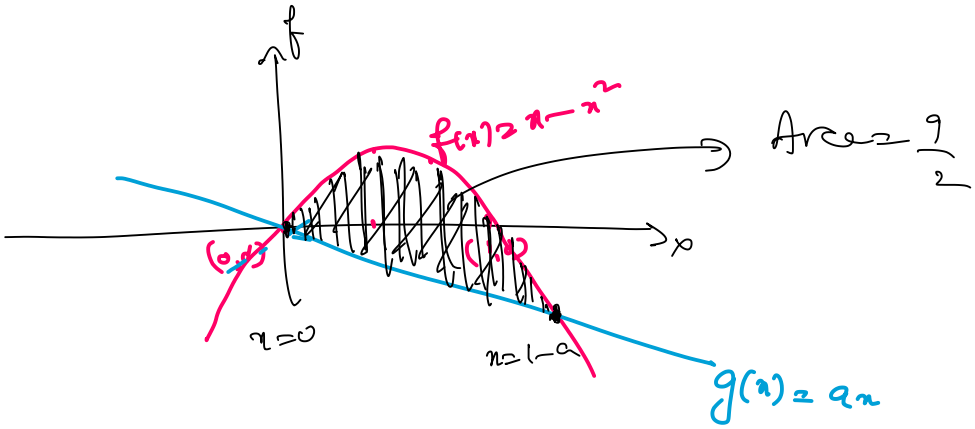
Ex: Find a function which is NOT integrable?

Applications of Integration:

Eg1: Let $f(x) = x - x^2$ and $g(x) = ax$.

Find 'a' so that the area of the region above graph of g and below graph of f

hey area $\frac{1}{2}$.



$$\text{Area} = \frac{9}{2}$$

$$\int_{x=0}^{1-a} (f(x) - g(x))\,dx = \frac{9}{2}$$

$$f(x) = g(x)$$

$$x - x^2 = ax$$

$$\Rightarrow x(1-x) = ax$$

$x = a$ $\quad$ $x = 1-a$

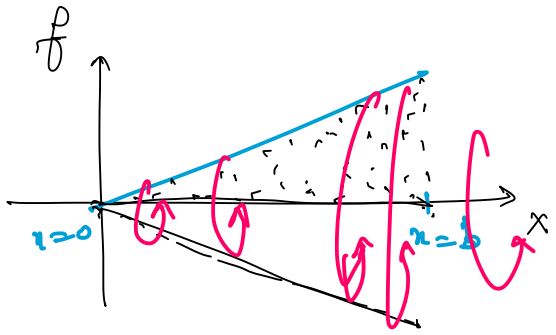
$$\Rightarrow \int_{x=0}^{1-a} (x - x^2 - ax)\,dx = \frac{9}{2}$$

$$\Rightarrow \frac{x^2}{2} - \frac{x^3}{3} - a\frac{x^2}{2}\Big|_{x=0}^{1-a} = \frac{9}{2}$$

$$\Rightarrow a = -2$$

Eg:) Find the volume of a solid generated by revolving the ordinate set of $f(x) = cx$, $c>0$, over +ve x-axis.

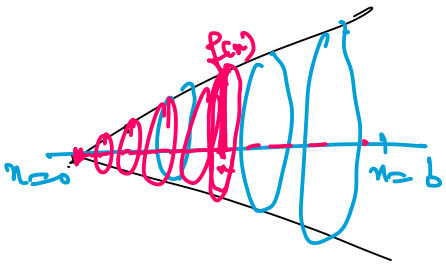
$(0 \le x \le b)$



ordinate set = $\{(x,y) : 0 \le x \le b,\ 0 \le y \le f(x)\}$

Volume of the solid

$$= \int_{x=0}^{b} \pi (f(x))^2 \, dx$$



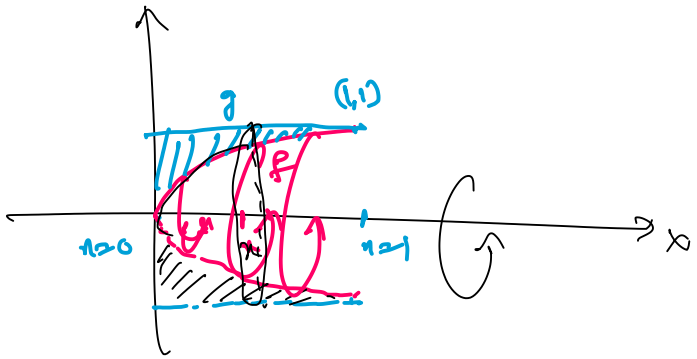
$$= \int_{x=0}^{b} \pi (cx)^2 \, dx$$

$$= \int_{x=0}^{b} \pi c^2 x^2 \, dx \qquad = \pi c^2 \cdot \frac{x^3}{3} \Big|_0^b$$

$$= \frac{\pi c^2 b^3}{3}.$$

Eg 1: Let $f(x) = \sqrt{x}$, $g(x) = 1$. $0 \le x \le 1$

Find the volume of the solid generated by revolving the region between f & g about +ve x-axis.



$$= \int_{x=0}^{1} \pi (f(x))^2 \, dx$$

$$+ \int_{x=0}^{1} \pi (g(x))^2 \, dx \qquad = \frac{2\pi}{3}$$

(Check!) ?